

Notes on Computer Class Exercises ST451 2025/2026

London School of Economics and Political Science

Computer Class 1

Activity 1. Assume that the posterior of the variable of interest θ , $\pi(\theta|y)$ is the $\text{Gamma}(2, 1)$ distribution. Provide the answers to the questions below using Monte Carlo. Verify by contrasting with the exact answers where possible.

- (a) Provide the Bayes estimate for θ .
- (b) Give a 95% credible interval for θ .
- (c) Calculate the posterior variance, $E[\theta^3|y]$ and $P(\theta > 3|y)$.
- (d) Give a kernel density plot for its pdf.

Activity 2. Assume that we observe 1 success in a Binomial experiment involving 7 trials. Assign $\text{Uniform}(0, 1)$ prior on the probability of success θ and derive its posterior. Then, provide the answers to the below using Monte Carlo. Verify by contrasting with the exact answers where possible.

- (a) Provide the Bayes estimate for θ .
- (b) Give a 95% credible interval for θ .
- (c) Calculate the posterior variance, $E[1/\theta|y]$ and $P(\theta < 0.2|y)$.
- (d) Give a kernel density plot for its pdf.

Activity 3. A bank manager wants to decide on whether she should employ additional staff in the cashiers of their branch. This is usually determined by the number of customers visiting their branch on an average day. **Sample:** An experiment is conducted and the number of customers visiting the branch is recorded on 10 random days and are shown below:

$$X = (86, 98, 77, 94, 107, 108, 93, 72, 89, 81)$$

- (a) Split the data into a training set (first 5 observations) and a test set (remaining observations).
- (b) Assign a $\mathcal{N}(\mu, 100)$ model and an improper prior to the training set. This gives a predictive distribution of the $\mathcal{N}(\bar{X}, \frac{100}{n} + 100)$. Use it to obtain point forecasts for the test set. Use both Frequentist and Bayesian inference and compare their performance.
- (c) Repeat the previous step but with 95% prediction intervals rather than point forecasts.
- (d) For the frequentist approach one can estimate the mean μ via the MLE, $\hat{\mu}$ which is the sample mean (of the training data) as with the Bayesian approach (only because we used an improper prior). The distribution for the future data can then be the $\mathcal{N}(\bar{X}, 100)$.
- (e) The MSE can be used to assess the forecasts. For the prediction intervals we can just count the number of test data contained in them.
- (f) Discuss the differences or similarities between the frequentist and Bayesian approaches.

Activity 1. Assume that the posterior of the variable of interest θ , $\pi(\theta|y)$ is the $\text{Gamma}(2, 1)$ distribution. Provide the answers to the questions below using Monte Carlo. Verify by contrasting with the exact answers where possible.

- (a) Provide the Bayes estimate for θ .
- (b) Give a 95% credible interval for θ .
- (c) Calculate the posterior variance, $E[\theta^3|y]$ and $P(\theta > 3|y)$.
- (d) Give a kernel density plot for its pdf.

(a) We can provide the median of θ under $\pi(\theta|y)$ as Bayes estimator. Recall the Gamma density with $(\text{shape}, \text{scale}) = (\alpha, \beta) = (2, 1)$ parametrisation (which, in general, is the one in `numpy.random.gamma`; check out the `scipy.stats.gamma` parametrisation):

$$\begin{aligned}\pi(\theta|y) &= \frac{1}{\Gamma(\alpha)\beta^\alpha} \theta^{\alpha-1} e^{-\theta/\beta} \\ &= \theta e^{-\theta}, \theta \geq 0\end{aligned}$$

The CDF is then

$$F_{\theta|y}(z) = \int_0^z \theta e^{-\theta} d\theta = 1 - (z+1)e^{-z}$$

which is continuous, strictly increasing (a bijection of $[0, \infty)$, $[0, 1)$) so the quantile function is

$$\Gamma_{2,1}(u) := F_{\theta|y}^{-1}(u) = \{u \in (0, 1) : (z+1)e^{-z} = 1 - u\}$$

We use `scipy.stats.gamma.ppf(q=u, a=2, loc=0, scale=1)` (note that $\beta = 1$ makes the parametrisation choice indifferent) to find the (almost exact) quantile value for given $u \in (0, 1)$.

(b) The 95% credible interval for the median is then $[\Gamma_{2,1}(0.025), \Gamma_{2,1}(0.975)]$ because:

$$\begin{aligned}P(\theta \in [\Gamma_{2,1}(0.025), \Gamma_{2,1}(0.975)]|y) \\ &= F_{\theta|y}(\Gamma_{2,1}(0.975)) - F_{\theta|y}(\Gamma_{2,1}(0.025)) \\ &= 0.975 - 0.025 = 0.95\end{aligned}$$

(c) We have

$$\begin{aligned}V[\theta|y] &= \alpha\beta^2 = 2, \quad E[\theta^3|y] = \int_0^\infty \theta^4 e^{-\theta} d\theta = \Gamma(5) = 24 \\ P(\theta > 3|y) &= 1 - F_{\theta|y}(3) = 4e^{-3} \approx 0.1991\end{aligned}$$

(d) Given a sample $X = (X_1, \dots, X_n)$ of length n drawn from $\pi(\theta|y)$, the KDE is

$$f(x) = \frac{1}{n} \sum_{1 \leq k \leq n} K(x - X_k), \quad x \in \mathbb{R}$$

for kernel $K \geq 0$ with $\|K\|_1 = 1$; in `scipy.stats.gaussian_kde` it is a normal density.

Activity 2. Assume that we observe 1 success in a Binomial experiment involving 7 trials. Assign Uniform(0, 1) prior on the probability of success θ and derive its posterior. Then, provide the answers to the below using Monte Carlo. Verify by contrasting with the exact answers where possible.

- (a) Provide the Bayes estimate for θ .
- (b) Give a 95% credible interval for θ .
- (c) Calculate the posterior variance, $E[1/\theta|y]$ and $P(\theta < 0.2|y)$.
- (d) Give a kernel density plot for its pdf.

(a) We derive the general posterior given uniform prior on θ given a vector $y \in \{0, 1\}^n$ with average $\bar{y}$:

$$\begin{aligned} f(\theta|y) &\propto f(y|\theta)f(\theta) \\ &= \binom{n}{n\bar{y}} \theta^{n\bar{y}} (1-\theta)^{n(1-\bar{y})} \cdot \mathbf{1}_{[0,1]}(\theta) \end{aligned}$$

Hence $f(\theta|y)$ is a Beta density with $(\alpha, \beta) = (n\bar{y} + 1, n + 1 - n\bar{y})$ because

$$\begin{aligned} \binom{n}{n\bar{y}} \int_{[0,1]} \theta^{n\bar{y}} (1-\theta)^{n(1-\bar{y})} d\theta &= \binom{n}{n\bar{y}} \frac{\Gamma(n\bar{y} + 1)\Gamma(n + 1 - n\bar{y})}{\Gamma(n + 2)} \\ &= \binom{n}{n\bar{y}} \frac{(n\bar{y})!(n(1 - \bar{y}))!}{(n + 1)!} = \frac{1}{n + 1} \end{aligned}$$

So

$$f(\theta|y) = \frac{(n + 1)!}{(n\bar{y})!(n(1 - \bar{y}))!} \theta^{n\bar{y}} (1 - \theta)^{n(1 - \bar{y})}$$

Here we have $(n, n\bar{y}) = (7, 1)$ so $(\alpha, \beta) = (2, 7)$. We can find the quantile at $u \in (0, 1)$ with `scipy.stats.beta.ppf(q=u, a=2, b=7)`. We can provide the median of θ under $f(\theta|y)$ as Bayes estimator.

(b) If $B_{2,7}(u)$ is the quantile function of Beta(2, 7) and $F_{\theta|y}$ is its CDF, we see that

$$\begin{aligned} P(\theta \in [B_{2,7}(0.025), B_{2,7}(0.975)]|y) \\ &= F_{\theta|y}(B_{2,7}(0.975)) - F_{\theta|y}(B_{2,7}(0.025)) \\ &= 0.975 - 0.025 = 0.95 \end{aligned}$$

(c) We get

$$\begin{aligned} V[\theta|y] &= \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \frac{14}{810} \\ E[1/\theta|y] &= \frac{(n + 1)!}{(n\bar{y})!(n(1 - \bar{y}))!} \int_{[0,1]} \theta^{n\bar{y}-1} (1 - \theta)^{n(1-\bar{y})} d\theta = \frac{n + 1}{n\bar{y}} = 8 \end{aligned}$$

For $P(\theta < 0.2|y)$, we may use `scipy.stats.beta.cdf(x=0.2, a=2, b=7)`.

(d) Again, we use `scipy.stats.gaussian_kde` for a Gaussian KDE.

Activity 3. A bank manager wants to decide on whether she should employ additional staff in the cashiers of their branch. This is usually determined by the number of customers visiting their branch on an average day. **Sample:** An experiment is conducted and the number of customers visiting the branch is recorded on 10 random days and are shown below:

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- (a) Split the data into a training set (first 5 observations) and a test set (remaining observations).
- (b) Assign a $\mathcal{N}(\mu, 100)$ model and an improper prior to the training set. This gives a predictive distribution of the $\mathcal{N}(\bar{X}, \frac{100}{n} + 100)$. Use it to obtain point forecasts for the test set. Use both Frequentist and Bayesian inference and compare their performance.
- (c) Repeat the previous step but with 95% prediction intervals rather than point forecasts.
- (d) For the frequentist approach one can estimate the mean μ via the MLE, $\hat{\mu}$ which is the sample mean (of the training data) as with the Bayesian approach (only because we used an improper prior). The distribution for the future data can then be the $\mathcal{N}(\bar{X}, 100)$.
- (e) The MSE can be used to assess the forecasts. For the prediction intervals we can just count the number of test data contained in them.
- (f) Discuss the differences or similarities between the frequentist and Bayesian approaches.

(b), (Assign a $\mathcal{N}(\mu, 100)$ model and an improper prior to the training set. This gives a predictive distribution of the $\mathcal{N}(\bar{X}, \frac{100}{n} + 100)$)

The improper prior for μ in $\mathcal{N}(\mu, \sigma^2)$ with σ^2 known is the constant $f(\mu) = 1, \mu \in \mathbb{R}$. We get for $y = (y_1, \dots, y_n) \in \mathbb{R}^n$ with average $\bar{y} = \frac{1}{n}y'1$:

$$\begin{aligned} f(\mu|y) &\propto f(y|\mu)f(\mu) \\ &\propto \exp\left(-\frac{1}{2\sigma^2} \sum_{1 \leq k \leq n} (y_k - \mu)^2\right) \\ &= \exp\left(-\frac{1}{2\sigma^2} \left(n\mu^2 + \sum_{1 \leq k \leq n} y_k^2 - 2\mu n\bar{y}\right)\right) \\ &= \exp\left(-\frac{1}{2\sigma^2} \left(n\mu^2 + n\bar{y}^2 - 2\mu n\bar{y}\right)\right) \exp\left(-\frac{1}{2\sigma^2} \left(\sum_{1 \leq k \leq n} y_k^2 - n\bar{y}^2\right)\right) \\ &\propto \exp\left(-\frac{(\mu - \bar{y})^2}{2\sigma^2(\frac{1}{n})}\right) \end{aligned}$$

hence the posterior is $\mathcal{N}(\bar{y}, \sigma^2/n)$. Recall that the posterior predictive for $y_{n+1} \in \mathbb{R}$ is

$$f(y_{n+1}|y) = \int_{\mathbb{R}} f(y_{n+1}|\mu)f(\mu|y)d\mu$$

We thus get the posterior predictive

$$\begin{aligned} f(y_{n+1}|y) &= \int_{\mathbb{R}} f(y_{n+1}|\mu) f(\mu|y) d\mu \\ &\propto \int_{\mathbb{R}} \exp\left(-\frac{(y_{n+1}-\mu)^2}{2\sigma^2} - \frac{(\mu-\bar{y})^2}{2\sigma^2(\frac{1}{n})}\right) d\mu \end{aligned}$$

This is a Gaussian convolution. Indeed, write

$$u(\nu) := \int_{\mathbb{R}} \exp\left(-\frac{(\nu - (\mu - \bar{y}))^2}{2\sigma^2} - \frac{(\mu - \bar{y})^2}{2\sigma^2(\frac{1}{n})}\right) d\mu, \nu \in \mathbb{R}$$

It can be proved (sketch: substitute $x = \mu - \bar{y}$, complete the squares, use Gaussian integral) that

$$u(\nu) \propto \exp\left(-\frac{\nu^2}{2\sigma^2(1 + \frac{1}{n})}\right)$$

Therefore

$$\begin{aligned} &\int_{\mathbb{R}} \exp\left(-\frac{(y_{n+1}-\mu)^2}{2\sigma^2} - \frac{(\mu-\bar{y})^2}{2\sigma^2(\frac{1}{n})}\right) d\mu \\ &= \int_{\mathbb{R}} \exp\left(-\frac{((y_{n+1}-\bar{y}) - (\mu-\bar{y}))^2}{2\sigma^2} - \frac{(\mu-\bar{y})^2}{2\sigma^2(\frac{1}{n})}\right) d\mu \\ &= u(y_{n+1} - \bar{y}) \\ &\propto \exp\left(-\frac{(y_{n+1} - \bar{y})^2}{2\sigma^2(1 + \frac{1}{n})}\right) \end{aligned}$$

But then

$$f(y_{n+1}|y) = \frac{1}{\sqrt{2\pi\sigma^2(1 + \frac{1}{n})}} \exp\left(-\frac{(y_{n+1} - \bar{y})^2}{2\sigma^2(1 + \frac{1}{n})}\right)$$

That is

$$X_{n+1}|(X_1, \dots, X_n) = y \sim \mathcal{N}\left(\bar{y}, \sigma^2\left(1 + \frac{1}{n}\right)\right)$$

as we wanted to show.

Computer Class 2

Linear Regression / Polynomial Fitting

We start with the polynomial curve fitting exercise seen in the lecture slides. The *truth* is the function $\sin(2\pi x)$ for $x \in [0, 1]$. We observe 10 points, equispaced in $[0, 1]$ - array $\mathbf{x}$, with $\mathcal{N}(0, 0.3^2)$ independent measurement error, array $\mathbf{y}$. The array $\mathbf{xg}$ contains a grid of 100 points in $[0, 1]$.

Activity 1. Fit polynomials of 3rd and 9th degrees to the previous data and provide their coefficients and their training mean squared error.

Ridge and Lasso Regression

We will use all powers of $\mathbf{x}$ up to 9 but we also include a penalty term on the squared value of the coefficients. The amount of penalisation is controlled by the parameter ' λ '. We will repeat the exercise under a more realistic setting:

- Generate more 100 noisy observations from the function $\sin(2\pi x)$ as before.
- Randomly select 10 of those. These will form the **training sample** that will be used to fit the models and obtain the regression coefficients.
- Set aside the remaining 90 observations. These will be the **test sample** that will be used to assess the predictions of each model that was estimated in the train sample.
- Contrasting those predictions to the real data will allow us to estimate the **test error** and assess the out of sample performance of each model.
- Monitor both the training and test errors for each model.

Activity 2. Find the order of polynomial that produces the smallest test MSE. Can you beat that with Ridge or Lasso Regression? In other words can you find penalty ' λ ' such that the Ridge or Lasso regression model produces an even smaller test MSE?

Real data example

Finally we will look at a real data example (`diabetes.data.txt`), some info on which is given below:

- 442 diabetes patients;
- 10 main variables, $\mathbf{x}_1, \dots, \mathbf{x}_{10}$: age, gender, body mass index, average blood pressure (map), and six blood serum measurements (tc, ldl, hdl, tch, ltg, glu);
- 45 interactions of the form $\mathbf{x}_j \mathbf{x}_k$;
- 9 quadratic effects of the form $\mathbf{x}_j^2$;
- Measure of disease progression taken one year later, $\mathbf{y}$.

Activity 3. Fit and calculate the test error of the Ridge and Lasso regression models. Try to find an optimal λ for them.

Linear Regression / Polynomial Fitting

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Activity 1. Fit polynomials of 3rd and 9th degrees to the previous data and provide their coefficients and their training mean squared error.

We have a model of noisy observations $Y = (Y_1, \dots, Y_n)$ of the form

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ \dots \\ Y_n \end{bmatrix} = \begin{bmatrix} \sin(2\pi x_1) \\ \sin(2\pi x_2) \\ \sin(2\pi x_3) \\ \dots \\ \sin(2\pi x_n) \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \dots \\ \varepsilon_n \end{bmatrix}, \quad \varepsilon_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 0.3^2)$$

(this is array $\mathbf{y}$) where

$$(x_1, x_2, x_3, \dots, x_n) = \left(0, \frac{1}{n-1}, \frac{2}{n-1}, \dots, \frac{n-2}{n-1}, 1\right)$$

and we set $n = 10$ (this is array $\mathbf{x}$). Let $k \in \mathbb{N}_0$ and define the Vandermonde matrix

$$X = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 & \dots & x_1^k \\ 1 & x_2 & x_2^2 & x_2^3 & \dots & x_2^k \\ 1 & x_3 & x_3^2 & x_3^3 & \dots & x_3^k \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 1 & x_n & x_n^2 & x_n^3 & \dots & x_n^k \end{bmatrix}$$

and we find (with `sklearn.linear_model.LinearRegression`) $\hat{\beta} \in \mathbb{R}^{k+1}$ which solves

$$\hat{\beta} = \operatorname{argmin}_{\beta \in \mathbb{R}^{k+1}} \|Y - X\beta\|_2^2$$

which is solved in closed form by the OLS argument (if $X'X$ is invertible):

$$\begin{aligned} \frac{d}{d\beta} \|Y - X\beta\|_2^2 &= \frac{d}{d\beta} (Y - X\beta)'(Y - X\beta) \\ &= \frac{d}{d\beta} (Y'Y - 2X'Y\beta + \beta'X'X\beta) \\ &= -2X'Y + (X'X + (X'X)')\beta \\ &= -2X'Y + 2X'X\beta \\ \implies \hat{\beta} &= (X'X)^{-1}X'Y \end{aligned}$$

The fitted values and the training MSE are respectively

$$\hat{Y} = X\hat{\beta}, \quad \text{MSE} = \frac{1}{n} \|Y - \hat{Y}\|_2^2$$

Ridge and Lasso Regression

We will use all powers of $\mathbf{x}$ up to 9 but we also include a penalty term on the squared value of the coefficients. The amount of penalisation is controlled by the parameter ' $\mathbf{lam}$ '. We will repeat the exercise under a more realistic setting:

- (a) Generate more 100 noisy observations from the function $\sin(2\pi x)$ as before.
- (b) Randomly select 10 of those. These will form the **training sample** that will be used to fit the models and obtain the regression coefficients.
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- (d) Contrasting those predictions to the real data will allow us to estimate the **test error** and assess the out of sample performance of each model.
- (e) Monitor both the training and test errors for each model.

Activity 2. Find the order of polynomial that produces the smallest test MSE. Can you beat that with Ridge or Lasso Regression? In other words can you find penalty ' $\mathbf{lam}$ ' such that the Ridge or Lasso regression model produces an even smaller test MSE?

Given $\lambda > 0$ (this is $\mathbf{lam}$), regression matrix X and observations Y , the Ridge regression coefficient found by the regression model

```
sklearn.linear_model.Ridge(alpha=lam)
```

solves

$$\hat{\beta}^{\text{Ridge}} = \operatorname{argmin}_{\beta \in \mathbb{R}^{k+1}} (\|Y - X\beta\|_2^2 + \lambda \|\beta\|_2^2)$$

which is found in closed form by the OLS argument (if $X'X + \lambda I_k$ is invertible):

$$\begin{aligned} \frac{d}{d\beta} (\|Y - X\beta\|_2^2 + \lambda \|\beta\|_2^2) &= \frac{d}{d\beta} ((Y - X\beta)'(Y - X\beta) + \lambda \beta' \beta) \\ &= \frac{d}{d\beta} (Y'Y - 2\tilde{X}'Y\beta + \beta'X'X\beta + \lambda \beta' \beta) \\ &= -2X'Y + (X'X + (X'X)')\beta + 2\lambda\beta \\ &= -2X'Y + 2X'X\beta + 2\lambda\beta \\ \implies \hat{\beta}^{\text{Ridge}} &= (X'X + \lambda I_k)^{-1} X'Y \end{aligned}$$

Given $\lambda > 0$ (again, this is $\mathbf{lam}$) the LASSO regression coefficient found by the regression model

```
sklearn.linear_model.Lasso(alpha=lam)
```

solves

$$\hat{\beta}^{\text{LASSO}} = \operatorname{argmin}_{\beta \in \mathbb{R}^{k+1}} \left(\|Y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right)$$

where m is the dimension of Y . Unlike $\hat{\beta}^{\text{Ridge}}$, the estimate $\hat{\beta}^{\text{LASSO}}$ is necessarily found numerically. In both cases, fitted values and MSE are found as in polynomial regression.

Computer Class 3

Automobile Bodily Injury Claims Data

The data are automobile injury claims data using data from the Insurance Research Council (IRC), a division of the American Institute for Chartered Property Casualty Underwriters and the Insurance Institute of America. The data, collected in 2002, contains information on the gender of the claimant, attorney involvement, years of driving experience and the economic loss (**LOSS**, in thousands). A detailed description of the variables in the data is provided below:

- (a) **Attorney**: Whether the claimant is represented by an attorney (= 1 if yes and = 0 if no);
- (b) **CLMSEX**: Claimant's gender (= 1 if male and = 0 if female);
- (c) **CLMAGE**: Claimant's age minus the age driving license was obtained;
- (d) **LOSS**: The claimant's total economic loss (in thousands \$).

For confidentiality issues we consider here a sample of simulated data similar to that of a state is considered. The data are in the provided file 'automobileBI.csv'.

Activity 1. Obtain the linear regression MLEs and 95% confidence intervals for the model with the **LOSS** variable as response with only the **ATTORNEY** variable and without using any Python built in function. Check your answers against the **statsmodels** corresponding function (`statsmodels.regression.linear_model.OLS`).

Bayesian Linear Regression

We now turn to Bayesian inference. We start by calculating the posterior parameters that correspond to the unit information prior. For σ^2 we set the prior $\text{IGamma}(a_0, b_0)$.

Activity 2. Consider all possible regression models with the **LOSS** variable as response and some (or all or none) of the other variables present (in their current form, no polynomials). There are 8 such models. Calculate the model evidence for each of them, report the optimal one and present posterior estimates of its coefficients.

Activity 3. Given that the scatter plots do not provide support for the linear model, consider transforming the target and/or the features and repeat.

Automobile Bodily Injury Claims Data

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- (a) **Attorney**: Whether the claimant is represented by an attorney (= 1 if yes and = 0 if no);
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For confidentiality issues we consider here a sample of simulated data similar to that of a state is considered. The data are in the provided file 'automobileBI.csv'.

Activity 1. Obtain the linear regression MLEs and 95% confidence intervals for the model with the LOSS variable as response with only the ATTORNEY variable and without using any Python built in function. Check your answers against the statsmodels corresponding function (statsmodels.regression.linear_model.OLS).

Given a linear model of the form for the response $Y = (Y_1, Y_2, \dots, Y_n)$

$$Y = X\beta + \varepsilon, \quad X = \begin{bmatrix} 1 & x_{1,1} & \dots & x_{1,m} \\ 1 & x_{2,1} & \dots & x_{2,m} \\ \dots & \dots & \dots & \dots \\ 1 & x_{n,1} & \dots & x_{n,m} \end{bmatrix}, \quad \varepsilon \sim \mathcal{N}_n(0, \sigma^2 I_n)$$

where $\beta \in \mathbb{R}^m$, the log-likelihood for (β, σ^2) is

$$\ell(y; \beta, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|y - X\beta\|_2^2, \quad y \in \mathbb{R}^n$$

Given any $\sigma^2 > 0$, the second term is always maximised by the OLS estimate $\hat{\beta} = (X'X)^{-1}X'y$. On the other hand,

$$\frac{d}{d\sigma^2} \ell(y; \hat{\beta}, \sigma^2) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \|y - X\hat{\beta}\|_2^2 = 0 \implies \hat{\sigma}^2 = \frac{1}{n} \|y - X\hat{\beta}\|_2^2$$

So the MLE is $(\hat{\beta}, \hat{\sigma}^2)$. Since Y is multivariate normal, we have that $\hat{\beta}$ is also multivariate normal. Thus

$$\begin{aligned} E[\hat{\beta}] &= (X'X)^{-1}X'E[Y] = \beta \\ V[\hat{\beta}] &= (X'X)^{-1}X'V[Y]X(X'X)^{-1} = \sigma^2(X'X)^{-1} \end{aligned}$$

So we may choose to obtain marginal confidence intervals for $\hat{\beta}$ from $\mathcal{N}_m(\hat{\beta}, \hat{\sigma}^2(X'X)^{-1})$ (if $m \ll n$), or from a t -distribution argument (if m is close to n).

Bayesian Linear Regression

We now turn to Bayesian inference. We start by calculating the posterior parameters that correspond to the unit information prior. For σ^2 we set the prior $\text{IGamma}(a_0, b_0)$.

Activity 2. Consider all possible regression models with the LOSS variable as response and some (or all or none) of the other variables present (in their current form, no polynomials). There are 8 such models. Calculate the model evidence for each of them, report the optimal one and present posterior estimates of its coefficients.

Given $\sigma^2 > 0$, the unit information prior is a multivariate normal prior distribution for β which depends on X . If $X'X$ is invertible, it is specified by (Murphy, p. 236)

$$\begin{aligned}\Omega_0 &:= n(X'X)^{-1} \\ \mu_0 &\in \mathbb{R}^m \\ \beta|X, \sigma^2 &\sim \mathcal{N}_m(\mu_0, \sigma^2 \Omega_0)\end{aligned}$$

We specify an $\text{IGamma}(a_0, b_0)$ as a prior for σ^2 . It can be proved (tedious, but straightforward) that the joint posterior density is thus (Murphy, 7.69-7.73 p. 235)

$$\begin{aligned}f(\beta, \sigma^2|X, y) &= f(y|X, \beta, \sigma^2)f(\beta|X, \sigma^2)f(\sigma^2) \\ &\propto \frac{1}{(\sigma^2)^{m/2}} \exp\left(-\frac{1}{2\sigma^2}(\beta - \mu_n)'(X'X + \Omega_0^{-1})(\beta - \mu_n)\right) \\ &\quad \cdot \frac{1}{(\sigma^2)^{\frac{n+2a_0}{2}+1}} \exp\left(-\frac{2b_0 + \|y\|_2^2 - \mu_n' \Omega_n^{-1} \mu_n + \mu_0' \Omega_0^{-1} \mu_0}{2\sigma^2}\right) \\ &\propto f(\beta|\sigma^2, X, y)f(\sigma^2|X, y)\end{aligned}$$

where

$$\begin{aligned}\beta|\sigma^2, X, y &\sim \mathcal{N}(\mu_n, \sigma^2 \Omega_n) \\ \sigma^2|X, y &\sim \text{IGamma}(a_n, b_n)\end{aligned}$$

and

$$\begin{aligned}\mu_n &:= \Omega_n(X'y + \Omega_0^{-1}\mu_0) \\ \Omega_n &:= (X'X + \Omega_0^{-1})^{-1} \\ a_n &:= a_0 + \frac{n}{2} \\ b_n &:= b_0 + \frac{1}{2}(\|y\|_2^2 - \mu_n' \Omega_n^{-1} \mu_n + \mu_0' \Omega_0^{-1} \mu_0)\end{aligned}$$

The model evidence (which depends only on X) is

$$\begin{aligned}f(y|X) &= \frac{f(y|X, \beta, \sigma^2)f(\beta, \sigma^2|X)}{f(\beta, \sigma^2|X, y)} \\ \implies \ln f(y|X) &= \underbrace{\ln f(y|X, \beta, \sigma^2)}_{\text{log-likelihood}} + \underbrace{\ln f(\beta, \sigma^2|X)}_{\text{log-prior}} - \underbrace{\ln f(\beta, \sigma^2|X, y)}_{\text{log-posterior}}\end{aligned}$$

We can evaluate the rhs for admissible values of β, σ^2 (e.g. $(\mu_n, \frac{b_n}{a_n-1})$ if $a_n > 1$, the respective means of the posteriors).

Computer Class 4

Breast Cancer Survival Data

The dataset contains cases from a study that was conducted between 1958 and 1970 at the University of Chicago's Billings Hospital on the survival of patients who had undergone surgery for breast cancer. Four columns:

- (a) **age**: age of patient at time of operation (numerical);
- (b) **year**: patient's year of operation (year - 1900, numerical);
- (c) **nodes**: number of positive axillary nodes detected (numerical);
- (d) **y**: survival status (class attribute): 1 = the patient survived 5 years or longer, 2 = the patient died within 5 years.

See more info at the UCI repository <https://archive.ics.uci.edu/ml/machine-learning-databases/haberman/haberman.names>.

Activity 1. Provide the output from the MLE logistic regression and Bayesian logistic regression (via Laplace approximation) approaches.

Activity 2. Consider 2 logistic regression models:

- (a) $M1$: with covariates **age** and **nodes**.
 - (b) $M2$: with covariates **year**, **age** and **nodes**.
- (a) Compare them using the log-evidence, the BIC, the area under the ROC curve and the log scoring rule. (b) Discuss the differences in predicted probabilities from MLE and Bayesian models.

Linear Discriminant Analysis

Compute the classification probability $P(Y = 1|X)$ by modeling the joint probability $P_{Y,X}$. That is called a generative model.

Activity 3. Fit logistic regression and LDA models on the training data and evaluate their predictive performance (ROC, area under the ROC, log score) on the test data. Use **year**, **age** and **nodes**.

Breast Cancer Survival Data

The dataset contains cases from a study that was conducted between 1958 and 1970 at the University of Chicago's Billings Hospital on the survival of patients who had undergone surgery for breast cancer. Four columns:

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See more info at the UCI repository <https://archive.ics.uci.edu/ml/machine-learning-databases/haberman/haberman.names>.

Activity 1. Provide the output from the [MLE logistic regression](#) and [Bayesian logistic regression \(via Laplace approximation\)](#) approaches.

Logistic regression is a generalised linear model (GLM) specialised to binary data $Y = (Y_1, \dots, Y_n)$ with $P(Y \in \{0, 1\}^n) = 1$ and regressor matrix $X \in \mathbb{R}^{n \times (m+1)}$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} = \begin{bmatrix} 1 & x_{1,1} & \dots & x_{1,m} \\ 1 & x_{2,1} & \dots & x_{2,m} \\ \dots & \dots & \dots & \dots \\ 1 & x_{n,1} & \dots & x_{n,m} \end{bmatrix}$$

The GLM is characterised by the logit link function g :

$$g(z) = \ln \frac{z}{1-z} \implies P(Y_k = 1|X) = g^{-1}(x_k \cdot \beta), \quad g^{-1}(y) = \frac{1}{1 + e^{-y}}$$

[MLE logistic regression](#) (Murphy 8.3.1-3 pp. 246-250, Bishop 4.3.2-4.3.3 pp. 205-208). The likelihood is given by

$$L(y|X, \beta) = \prod_{1 \leq k \leq n} (g^{-1}(x_k \cdot \beta))^{y_k} (1 - g^{-1}(x_k \cdot \beta))^{1-y_k}, \quad y \in \{0, 1\}^n$$

and we aim to find $\beta = (\beta_0, \dots, \beta_m)$ which minimises (resp. maximises) the negative log-likelihood $-\ell = -\ln L$ (resp. the log-likelihood $\ell = \ln L$). It can be shown that (recall x_k are row vectors):

$$\begin{aligned} \nabla(-\ell)(\beta) &= \begin{bmatrix} \partial_{\beta_0}(-\ell)(\beta) \\ \dots \\ \partial_{\beta_m}(-\ell)(\beta) \end{bmatrix} = \sum_{1 \leq k \leq n} (g^{-1}(x_k \cdot \beta) - y_k) x'_k \\ H(-\ell)(\beta) &= \begin{bmatrix} \partial_{\beta_0^2}(-\ell)(\beta) & \dots & \partial_{\beta_0 \beta_m}(-\ell)(\beta) \\ \dots & \dots & \dots \\ \partial_{\beta_0 \beta_m}(-\ell)(\beta) & \dots & \partial_{\beta_m^2}(-\ell)(\beta) \end{bmatrix} = \sum_{1 \leq k \leq n} g^{-1}(x_k \cdot \beta)(1 - g^{-1}(x_k \cdot \beta)) x'_k x_k \end{aligned}$$

It can also be shown that $H_\beta(-\ell)$ is convex, hence $-\ell$ has a unique global minimum. Since no closed form is known, a naïve algorithm of Newton-Raphson type to find a minimum may follow the pseudocode:

Initialise $\beta = \beta^{(0)}$.

For $k = 1, 2, \dots$ **until convergence do**:

- (a) **Evaluate** $\nabla(-\ell)(\beta^{(k-1)})$;
- (b) **Evaluate** $H(-\ell)(\beta^{(k-1)})$;
- (c) **Set** $\beta^{(k)} = \beta^{(k-1)} - (H(-\ell)(\beta^{(k-1)}))^{-1} \nabla(-\ell)(\beta^{(k-1)})$

where we need to specify a **convergence** condition. The estimate $\hat{\beta}$ can be chosen to be the value obtained by the algorithm, if convergence is reached. In general, the library `scikit-learn` provides the logistic regression class

`sklearn.linear_model.LogisticRegression`

which can use a variety of robust solver algorithms to estimate β .

[Bayesian logistic regression \(via Laplace approximation\)](#) (Murphy 8.4.1-4 pp. 255-260). Suppose that β^M is the (real) modal value of the posterior density of β , and that the posterior has the form

$$f(\beta|X, y) \propto \exp(-\mathcal{E}(\beta))$$

$$\mathcal{E}(\beta) \approx \mathcal{E}(\beta^M) + (\beta - \beta^M) \underbrace{\nabla(\mathcal{E})(\beta^M)}_{=0} + \frac{1}{2}(\beta - \beta^M)' H(\mathcal{E})(\beta^M)(\beta - \beta^M)$$

so that we obtain the Laplace approximation

$$\beta|X, y \sim \mathcal{N}(\beta^M, (H(\mathcal{E})(\beta^M))^{-1})$$

So we may start out with a Gaussian prior $\beta \sim \mathcal{N}(\beta_0, \Sigma_0)$, obtaining

$$f(\beta|X, y) = L(y|X, \beta)f(\beta)$$

and we set the mean (and mode) of the Gaussian posterior as the MAP estimate

$$\hat{\beta}^M := \operatorname{argmax}_{\beta} f(\beta|X, y)$$

and the precision matrix (inverse of variance-covariance matrix) as

$$\begin{aligned} H(\mathcal{E})(\hat{\beta}^M) &= H(-\ln f(\beta|X, y))(\hat{\beta}^M) \\ &= -H(\ln f(\beta))(\hat{\beta}^M) + H(-\ln L(y|X, \beta))(\hat{\beta}^M) \\ &= \Sigma_0^{-1} + H(-\ell)(\hat{\beta}^M) \end{aligned}$$

Again, in order to find $\hat{\beta}^M$ we can implement a naïve Newton-Raphson algorithm to minimise the negative log-likelihood.

Activity 2. Consider 2 logistic regression models:

- (a) $M1$: with covariates `age` and `nodes`.
- (b) $M2$: with covariates `year`, `age` and `nodes`.

(a) Compare them using the [log-evidence](#), the [BIC](#), the [area under the ROC curve](#) and the [log scoring rule](#). (b) Discuss the differences in predicted probabilities from MLE and Bayesian models.

[Log-evidence](#). (seen previously)

[BIC \(Bayesian information criterion\)](#). Defined as

$$\text{BIC} := \underbrace{\ell(\hat{\beta})}_{\text{log-likelihood at } \beta \text{ estimate}} - \underbrace{\frac{m}{2} \ln n}_{\text{penalty on number of regressors}}$$

As a model choice criterion, the higher BIC is preferable.

[Area under the ROC \(receiver operating characteristic\) curve](#). A binary classifier determines the class of $Y \in \{0, 1\}$ given X by comparing $P(Y = 1|X)$ with a threshold τ , that is

$$\hat{Y} = 1 \iff P(Y = 1|X) > \tau$$

Given a test set, the true positive rate (TPR) and the false positive rate (FPR) given a threshold τ are defined as

$$\begin{aligned} \text{TPR}(\tau) &:= \frac{\text{correct classifications as class 1}}{\text{total observations with class 1}} \\ \text{FPR}(\tau) &:= \frac{\text{incorrect classifications as class 1}}{\text{total observations with class 0}} \end{aligned}$$

The estimated ROC curve is the graph $\varphi(\tau) = (\text{FPR}(\tau), \text{TPR}(\tau))$ in $[0, 1]^2$ for some values of τ , and the area under the ROC curve is the estimated area of the unit square below the ROC curve. From `scikit-learn` we use

```
sklearn.metrics.roc_curve(y_true, y_score)
sklearn.metrics.roc_auc_score(y_true, y_score)
```

where `y_score` are the model predictions for `y_true`. As a model choice criterion, the higher area under the ROC curve (AUC) is preferable.

[Log-scoring rule \(or log-loss\)](#). Defined as

$$\text{LS}(Y, X) := -\ln \left(P(Y = 1|X)^Y (1 - P(Y = 1|X))^{1-Y} \right)$$

Given two models with respective $P(Y = 1|X)$, the lower LS is preferable. Given a test set, we use

```
sklearn.metrics.log_loss(y_true, y_pred)
```

where `y_pred` are the model predictions for `y_true`.

Linear Discriminant Analysis

Compute the classification probability $P(Y = 1|X)$ by modeling the joint probability $P_{Y,X}$. That is called a generative model.

Activity 3. Fit logistic regression and LDA models on the training data and evaluate their predictive performance (ROC, area under the ROC, log score) on the test data. Use `year`, `age` and `nodes`.

We shall use the classification model

`sklearn.discriminant_analysis.LinearDiscriminantAnalysis`

which implements Gaussian linear discriminant analysis (Gaussian LDA) (Murphy 4.2.2-4.2.4 pp. 103-106). We thus assume a Bernoulli marginal with parameter π for the binary class Y and Gaussian conditional $X|Y$ of the regressor $X = (X_1, \dots, X_m)$ which different means μ_0, μ_1 for the two classes and identical variance-covariance matrix Σ :

$$\begin{aligned} Y &\sim \text{Bernoulli}(\pi) \\ X|Y = k &\sim \mathcal{N}_m(\mu_k, \Sigma), \quad k = 0, 1 \end{aligned}$$

which yields for $x \in \mathbb{R}^m$

$$\begin{aligned} P(Y = 1|X = x) &= \frac{f(x|Y = 1)\pi}{f(x|Y = 1)\pi + f(x|Y = 0)(1 - \pi)} \\ &= \frac{1}{1 + \frac{f(x|Y=0)P(Y=0)}{f(x|Y=1)P(Y=1)}} \end{aligned}$$

Now note:

$$\begin{aligned} &\frac{f(x|Y = 0)P(Y = 0)}{f(x|Y = 1)P(Y = 1)} \\ &= \frac{1 - \pi}{\pi} \exp \left(-\frac{1}{2}(x'\Sigma^{-1}x + \mu_0'\Sigma^{-1}\mu_0 - 2\mu_0'\Sigma^{-1}x) + \frac{1}{2}(x'\Sigma^{-1}x + \mu_1'\Sigma^{-1}\mu_1 - 2\mu_1'\Sigma^{-1}x) \right) \\ &= \exp \left(-\underbrace{\left(\ln \frac{\pi}{1 - \pi} + \frac{1}{2}(\mu_0'\Sigma^{-1}\mu_0 - \mu_1'\Sigma^{-1}\mu_1) \right)}_{=:C} - \underbrace{(\mu_1 - \mu_0)'\Sigma^{-1}x}_{=: \beta'x} \right) \\ &= \exp(-(C + \beta'x)) \end{aligned}$$

Hence:

$$P(Y = 1|X = x) = \frac{1}{1 + e^{-(C + \beta'x)}}$$

If we set `solver='lsqr'` then the estimates $\hat{\pi}, \hat{\mu}_0, \hat{\mu}_1, \hat{\Sigma}$ are the MLE estimates of the Gaussian LDA.

Computer Class 5

Mean Field Approximation

Activity 1. Let $Y = (Y_1, \dots, Y_n)$ be independent $\text{Poisson}(\lambda)$ observations. Assume that λ follows the $\text{Gamma}(2, \beta)$ distribution, where β follows the $\text{Exponential}(1)$ distribution. The aim is to draw inference from the posterior $\pi(\theta|y)$, where $\theta = (\lambda, \beta)$.

The variational Bayes algorithm approximates $\pi(\theta|y)$ using the mean field approximation

$$q(\theta|y, \phi) = q(\lambda|y, \phi)q(\beta|y, \phi)$$

It can be shown that such an algorithm may consist of the following steps:

- (a) Initialise at $q(\lambda)$ to be the $\text{Gamma}(a_\lambda, b_\lambda)$ and $q(\beta)$ to be the $\text{Gamma}(a_\beta, b_\beta)$ distribution, setting

$$a_\lambda = 2 + \sum_i Y_i, \quad b_\lambda = b_\lambda^0, \quad a_\beta = 3, \quad \text{and} \quad b_\beta = b_\beta^0.$$

- (b) Iteratively update b_λ and b_β until the parameters or the ELBO converge. At iteration i will have:

- (i). Set

$$b_\lambda^i = n + E_{q(\beta)}[\beta] = n + 3/b_\beta^{i-1}$$

- (ii). Set

$$b_\beta^i = 1 + E_{q(\lambda)}[\lambda] = 1 + \left(2 + \sum_i Y_i\right) / b_\lambda^{i-1}$$

Task: Code the above algorithm and fit it to the simulated data. Check your answers in terms of the lambda estimates.

Automatic Differentiation Variational Inference (ADVI) with PyMC

Demonstration in Logistic (Binomial) Regression

$$\mu_k = \beta_0 + \beta_1 x_k, \quad x_k \in [-10, 20]$$

$$p_k = \frac{1}{1 + e^{-\mu_k}}$$

$$Y_k \sim \text{Binomial}(n, p_k), \quad \text{independent}$$

Activity 2. In order to incorporate a deterministic function in the PyMC model definition, you can use the function below (for the sigmoid function)

```
p = pymc.Deterministic("p", pm.math.invlogit(mu))
```

Also, the Bernoulli likelihood can be specified with the code below (with `n`, `p` and `observed` to be provided by the user):

```
pymc.Binomial("y", n=..., p=..., observed= ...)
```

Mean Field Approximation

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The variational Bayes algorithm approximates $\pi(\theta|y)$ using the mean field approximation

$$q(\theta|y, \phi) = q(\lambda|y, \phi)q(\beta|y, \phi)$$

It can be shown that such an algorithm may consist of the following steps: (...)

(shape-rate Gamma!) The log-joint distribution is characterised by the log-density

$$\begin{aligned} \ln f(y, \lambda, \beta) &= \ln \left(\prod_{1 \leq k \leq n} \frac{e^{-\lambda} \lambda^{y_k}}{y_k!} \right) + \ln \left(\frac{\beta^2}{\Gamma(2)} \lambda e^{-\beta \lambda} \right) + \ln(e^{-\beta}) \\ &= \left(-n\lambda + \ln \lambda \sum_{1 \leq k \leq n} y_k \right) + (2 \ln \beta + \ln \lambda - \lambda \beta) + (-\beta) + C^{(1)} \end{aligned}$$

For $\ln q(\lambda)$, we focus on the terms involving λ :

$$\begin{aligned} \ln q(\lambda) &= -(n + E_{q(\beta)}[\beta])\lambda + \ln \lambda \left(\sum_{1 \leq k \leq n} y_k + 1 \right) + C^{(2)} \\ \implies q(\lambda) &\propto e^{-\lambda(n + E_{q(\beta)}[\beta])} \lambda^{\sum_{1 \leq k \leq n} y_k + 1} \\ \implies \lambda &\sim \text{Gamma} \left(2 + \sum_{1 \leq k \leq n} y_k, n + E_{q(\beta)}[\beta] \right) \end{aligned}$$

while for $\ln q(\beta)$ we focus on the terms involving β :

$$\begin{aligned} \ln q(\beta) &= 2 \ln \beta - \beta(1 + E_{q(\lambda)}[\lambda]) + C^{(3)} \\ \implies q(\beta) &\propto e^{-\beta(1 + E_{q(\lambda)}[\lambda])} \beta^2 \\ \implies \beta &\sim \text{Gamma} \left(3, 1 + E_{q(\lambda)}[\lambda] \right) \end{aligned}$$

If we initialise at $q(\lambda)$ to be the $\text{Gamma}(a_\lambda, b_\lambda)$ and $q(\beta)$ to be the $\text{Gamma}(a_\beta, b_\beta)$ distribution, with

$$a_\lambda = 2 + \sum_{1 \leq k \leq n} y_k, \quad b_\lambda = b_\lambda^0, \quad a_\beta = 3, \quad \text{and} \quad b_\beta = b_\beta^0$$

We get the desired recursion (recall that the Gamma average is shape/rate):

$$\begin{aligned} b_\lambda^i &= n + E_{q(\lambda)}[\beta] = n + \frac{3}{b_\beta^{i-1}} \\ b_\beta^i &= 1 + E_{q(\beta)}[\lambda] = 1 + \frac{2 + \sum_{1 \leq k \leq n} y_k}{b_\lambda^i} \end{aligned}$$